

MOHIT CHAMOLI

Graphic Era Hill University, Dehradun, Uttarakhand

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Summary

Driven Computer Science engineering student and CTO at PixelPulse with strong expertise in full-stack development, low-level systems, and machine learning. Proven track record of leading teams and architecting scalable applications. Passionate about solving complex problems through optimized algorithms and innovative AI solutions.

Education

Graphic Era Hill University, Dehradun

B.Tech in Computer Science & Engineering

Aug 2024 – Expected June 2028

CGPA: 8.55/10.0

Experience

PixelPulse

Feb 2026 – Present

Chief Technology Officer (CTO)

- Directing a 10+ member engineering team delivering 5+ client-facing web and mobile applications, overseeing architecture decisions, code quality standards, and project timelines.
- Defining technology stack selection and system architecture across 7 concurrent projects spanning React.js, Node.js, Flutter, and cloud-based deployments.
- Establishing engineering workflows including Git-based version control practices, code review processes, and CI/CD deployment pipelines, improving deployment efficiency by 30%.

Projects

Low-Level Systems Projects — C, Memory Management, Garbage Collection

- Designed a custom memory allocator in C with explicit free-list management, block splitting, coalescing, and boundary tag metadata for efficient dynamic memory allocation.
- Constructed a mark-and-sweep garbage collector supporting root set scanning, recursive heap traversal, and automatic reclamation of unreachable objects, reducing memory leaks by 100%.

BingeBuddy — React.js, Node.js, Socket.IO, WebSockets

- Created a real-time synchronized watch-party platform with concurrent room-based sessions and distributed playback state management for up to 50 simultaneous clients.
- Architected an event-driven system using Socket.IO for low-latency bidirectional communication, handling 1,000+ concurrent events and resolving synchronization conflicts.

CypherFusion — Python, AES, ECC, Cryptography

- Engineered a hybrid encryption system combining AES symmetric encryption with ECC-based asymmetric key exchange for robust end-to-end data security.
- Integrated the ECDH key agreement protocol, enabling secure session key derivation without transmitting private keys, significantly enhancing cryptographic resilience.

AfwahTracker — Python, NLP, Machine Learning, scikit-learn

- Developed a misinformation detection engine using NLP pipelines for text preprocessing, feature extraction, and supervised classification of 5,000+ news articles.
- Applied TF-IDF vectorization with machine learning classifiers for fake news detection, evaluating performance with precision, recall, and F1 metrics to attain a 92% accuracy rate.

Obscura — Flutter, Dart, WireGuard, Networking

- Launched a cross-platform open-source VPN application integrating the WireGuard protocol for encrypted network tunnel management and secure traffic routing.
- Configured low-level networking interfaces bridging WireGuard's cryptographic tunnel with the Flutter UI for real-time connection state monitoring.

GymQuest — Java, OOP, Design Patterns, Java Swing

- Programmed a gamified fitness tracking application applying MVC architecture, Observer, and Strategy design patterns with persistent state management to track 10+ daily user metrics.

Technical Skills

- Languages:** C, C++, Java, Python, JavaScript, SQL, Dart
- Systems Programming:** Custom Memory Allocator (C), Mark-and-Sweep Garbage Collector
- Backend:** Node.js, Express.js, FastAPI, REST APIs, JWT Authentication, Socket.IO
- Frontend:** React.js, HTML/CSS, Tailwind CSS, Vite
- AI/ML:** NLP, Machine Learning, spaCy, Sentence Transformers, Gemini API, scikit-learn
- Databases:** MongoDB, MySQL, Firebase Firestore
- Coursework:** Data Structures & Algorithms, Compiler Design, DBMS, Operating Systems, OOP

Achievements

- Achieved a LeetCode contest rating of 1,655 (top 17% globally) with 200+ DSA problems solved across arrays, trees, graphs, and dynamic programming.
- Maintained 1,500+ active contributions on GitHub with consistent open-source development across multiple repositories.